Ashish Singh

U.S. Citizen | (530) 383-8851 | ashish.530singh@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of California, Los Angeles (UCLA) Dec 2026
M.S., Applied Statistics and Data Science Los Angeles, California

University of California, Los Angeles (UCLA) Jul 2024
B.S., Statistics and Data Science Los Angeles, California

SKILLS

- **Programming & Querying:** Python (pandas, NumPy, scikit-learn, XGBoost, PyTorch, matplotlib), R (tidyverse, ggplot2), SQL (PostgreSQL, SQL Server)
- **Statistics & Machine Learning:** GLMs, elastic-net and cumulative-logit models, tree ensembles, mixed models, Bayesian methods, neural networks, calibration, conformal prediction, bootstrap, experimental design
- **Data & Tools:** ETL, data cleaning, Jupyter, Git/GitHub, AWS S3, Hugging Face, LLM APIs (OpenAI, Anthropic, Gemini), Tableau, Power BI, Excel, Blackbaud CRM, Jira

EXPERIENCE

UCLA Alumni Association Aug 2026 — Present
Data Processing and Analysis Intern Los Angeles, California (Remote)

- Resolved 32 Jira tickets end to end, one per alumni event or report, cleaning raw Excel submissions with Python scripts in Spyder and loading hundreds of attendee records into Blackbaud CRM through mass-add imports.
- Pulled alumni data from the university data warehouse with the team's shared Python reporting package, fixing Python and library version conflicts so the scripts ran reliably for departments across campus.
- Calculated alumni engagement metrics in Excel and designed department reports in Canva to UCLA brand standards, keeping Smartsheet trackers current so event hosts could follow each submission.

VectorSurv (UC Davis) Aug 2024 — May 2025
Research Assistant Davis, California

- Collaborated with a postdoctoral researcher and analytics team to analyze 8-band satellite imagery (2020 to 2023) in R, producing an environmental risk assessment that flagged high-risk areas for mosquito-borne disease exposure.
- Designed and implemented preprocessing pipelines in R (raster, tidyverse) to extract, clean, and structure large remote-sensing datasets, speeding up downstream analysis.
- Performed geospatial and multispectral analysis and built ggplot2 visualizations of spectral trends, identifying locations suitable for mosquito habitats to support targeted surveillance.

PROJECTS

M.S. Thesis: Ordinal Prediction of Chest Radiograph Urgency Sep 2026 — Present

- Predicting a three-level urgency label (normal, chronic or incidental only, acute) for chest radiographs in the Indiana University chest X-ray collection from biomedical vision-language model embeddings.
- Fitting penalized cumulative-logit models on a 70/15/15 split, with calibration checks and conformal prediction sets that quantify the uncertainty of each prediction.

The 99-Cent Question: Kalshi Prediction Markets (STATS 414) Oct 2026

- With a 3-person team, built a Python pipeline that pulled 7.17M hourly price candles across 839,613 markets from PostgreSQL and S3 into Parquet, then defined a study set of 10,351 favorites priced at 90 to 99 cents.
- Tuned elastic-net logistic regression, random forest, and XGBoost models on date-based splits; none beat the quoted price, which had the best test Brier score (0.04624).
- Found Economics favorites lost only 0.5% of the time versus 3.5% implied (event-level bootstrap 95% CI: -3.4 to -2.6 points).

Pneumonia Detection with LLM and Deep Learning Jan 2026 — Mar 2026

- Fine-tuned a large language model with CNN-based feature extraction in PyTorch to classify chest X-rays for pneumonia, using torchvision augmentation and normalization to improve robustness.

Predicting Heart Failure Mortality for Actuarial Risk Jan 2024 — Mar 2024

- Modeled mortality on a 368-patient, 60-predictor clinical dataset in Python, using VIF to reduce multicollinearity to 7 key predictors.
- Compared Logistic Regression, SGD SVM, and Random Forest with GridSearchCV; Random Forest reached 98.2% accuracy and 0.99 ROC AUC.